David Dwyer

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Education

Texas A&M University Aug 2026 – May 2030
General Engineering – Bachelor of Science in Mechanical Engineering (Intended) *College Station, TX*

Engineering Experience

Texas A&M Formula SAE EV – Chassis Subteam Sep 2026 – Present
Apprentice *College Station, TX*

Torsional Rigidity Test Jig | *SolidWorks* *In fabrication*

- Designed from scratch to measure the AME26 chassis's torsional stiffness and validate its FEA model
- Anchored the rear to concrete through water-jetted, welded steel mock uprights, with the suspension made rigid
- Mounted the front on a center pivot with arms past the mock uprights, so hanging weights apply a known torque

Brake Pedal Assembly | *Fusion 360, Static FEA* *Subteam design challenge*

- Designed a 6:1 floor-pivot pedal with a balance bar, twin master cylinders, BOTS, and BSPD pressure sensing
- Ran FEA at the 2,000 N rules load, refining boundary conditions and mesh to a factor of safety above 4

Projects

Custom Go-Kart | *Frame Fabrication, Powertrain, Aluminum Fabrication* Feb 2025

- Fabricated a go-kart frame from scratch around a 6.6 HP (224cc) Predator engine and custom-fitted CVT
- Machined aluminum engine mounts for tight packaging; tuned the powertrain with a new exhaust and carburetor
- Integrated chain drive, axles, brakes, and steering; validated a 50 MPH top speed, then engineered a roll cage

Scorpius – FTC Competition Robot | *Fusion 360, CNC Machining, TPU 3D Printing* Aug 2025 – Jun 2026

- Modeled full robot subassemblies in Fusion 360, running clearance and stress checks before release to production
- Prototyped a fully modular robot architecture with interchangeable subsystems for rapid repair and iteration
- Specialized in compliant features 3D-printed from TPU, tuning part geometry for controlled flex and durability
- Machined parts on a 3-axis CNC and integrated carbon fiber, aluminum, and 3D-printed parts into a light chassis

Technical Skills

CAD / CAE / CAM: SolidWorks, Onshape, Fusion 360, Static FEA, Computer-Aided Manufacturing (CAM)
Fabrication: 3-Axis CNC Machining, 3D Printing, Carbon Fiber Layup, Aluminum Fabrication, Hand & Power Tools
Languages & Software: Java, Python, Microsoft Office, Google Suite
Certifications: American Lifeguard Association Lifeguard & AED/CPR Certified

Leadership / Extracurricular

FIRST Tech Challenge Robotics – Teams 6832 & 3734 Sep 2022 – May 2026
Team Captain & Design Lead (2024–2026), Member (2022–2024) *Dallas, TX*

- Led a 15-member team to Regional and State Championships twice across design, build, code, and outreach
- Directed design of 3 robots totaling 600+ custom 3D-printed, carbon fiber, aluminum, and CNC-machined parts
- Designed and released an open-source chassis architecture actively adopted by 12+ FTC robotics teams worldwide
- Earned **1st Place Design Award** (2025, 2026), **1st Place Sustain Award**, and **2nd Place Inspire Award**
- Secured ∼$13,000 in industry sponsorships by pitching corporate partners including Southwest Airlines

Boy Scouts of America – Troop 838 Apr 2025
Eagle Scout, Project Manager & Instructor *Dallas, TX*

- Engineered and constructed rainwater collection platforms and navigation infrastructure for the community
- Managed a $2,100+ budget and directed 25+ volunteers from planning through install; featured in Dallas news

Work Experience

Glen Cove Swim Club Apr 2025 – Present
Pool Technician, Shift Leader, Lifeguard *Dallas, TX*

- Diagnosed and repaired filtration, chemical balancing, and safety equipment to reduce facility downtime
- Supervised staff during daily operations; earned **Rookie Lifeguard of the Year (2025)** for leadership